Total current $I = \frac{V}{R} = \frac{60}{6}$

$I = 10\ A$

Branch current $I_1 = \frac{V}{R_1} = \frac{60}{10}$

$I_1 = 6\ A$

Branch current $I_2 = \frac{V}{R_2} = \frac{60}{15}$

$I_2 = 4\ A$

$I = I_1 + I_2$

$10A = 6A + 4A$

$10\ A = 10\ A$

The sum of the branch currents is equal to the total current in a parallel circuit. Hence, the theorem is proved.



In parallel connection, the total resistance is always lower than the lowest resistance connected in the circuit.

Ex., If two resistors 10Ω and 15Ω,are connected in parallel circuit, the total resistance is lower than 10 Ω. R = 10 X 15 /25 = 6 Ω. Notice that the total resistance (6 Ω) is lower than the lowest resistance (10 Ω). (Here, 10 Ω is the lowest resistance connected in the circuit).

2 Two resistors of values 50 Ω and 50Ω are connected in parallel with 250 V power supply. Prove that the sum of the branch currents is equal to the total current.

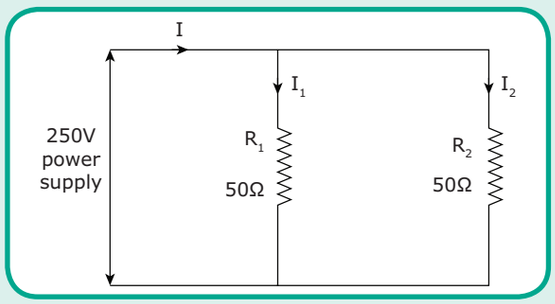


Figure 1.25

Total Resistance

$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{50 \times 50}{50 + 50} = \frac{2500}{100} = 25\Omega$

$R = 25\ \Omega$

Total current $I = \frac{V}{R} = \frac{250}{25} = 10A$

$I = 10\ A$

Branch current $I_1 = \frac{V}{R_1} = \frac{250}{50} = 5A$

$I_1 = 5\ A$

Branch current $I_2 = \frac{V}{R_2} = \frac{250}{50} = 5A$

$I_2 = 5\ A$

$I = I_1 + I_2$

$10A = 5A + 5A$

$10A = 10A$

The sum of the branch currents is equal to the total current in a parallel circuit. Hence the theorem is proved.

Activity 1.2

If two equal value resistors are connected in parallel circuit, the total resistance of the circuit will be half of any one of resistance value.

Ex., If 100 Ω and 100 Ω are connected in parallel, the total resistance is (100 X 100) / (100+100) = 10,000/200 = 50 Ω.

Shortly, 100/2 = 50 Ω.

Check it through various examples on your own.

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